

Practical 08

Aim : A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code :

```
GNU nano 8.7.1 LRU.c
#include <stdio.h>

int main() {
    int pages[12] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frame[3], time[3];
    int i,j,k,least;
    int faults = 0, hits = 0, counter = 0;

    for(i=0;i<3;i++)
        frame[i] = -1;

    for(i=0;i<12;i++){
        int found = 0;
        for(j=0;j<3;j++){
            if(frame[j] == pages[i]){
                hits++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }
        if(!found){
            least = 0;
            for(j=1;j<3;j++){
                if(time[j] < time[least])
                    least = j;
            }
            frame[least] = pages[i];
            counter++;
            time[least] = counter;
            faults++;
        }
    }

    printf("Page Hits = %d\n", hits);
    printf("Page Faults = %d\n", faults);
    printf("Hit Percentage = %.2f%%\n", (hits/12.0)*100);
    printf("Fault Percentage = %.2f%%\n", (faults/12.0)*100);

    return 0;
}
```

Output :

```
M ~
PRACHI@LAPTOP-CVJL7P0G UCRT64 ~
$ nano LRU.c

PRACHI@LAPTOP-CVJL7P0G UCRT64 ~
$ gcc LRU.c -o LRU

PRACHI@LAPTOP-CVJL7P0G UCRT64 ~
$ ./LRU
Page Hits = 0
Page Faults = 12
Hit Percentage = 0.00%
Fault Percentage = 100.00%

PRACHI@LAPTOP-CVJL7P0G UCRT64 ~
$ |
```

